TractX: A Density Aware Streamline Exclusion System for Accurate White Matter Reconstruction

Krishna Bhatt (BISV Grade 9) | Synopsys Science and Engineering Fair '24

VISUAL ABSTRACT

Issues and Drawbacks

Bottlenecks
Multiple fibers converge to a voxel with similar orientation.

Current Solutions
CSD, QBI, etc. account for crossing fiber issue; tractograms still have around 70% false positive rate.

Hypothesis
Utilizing fiber density as a marker for associated streamlines will reduce false positive rate.

Methodology

Identification
"Multi-bundle fixels" or fixels assigned to multiple bundles.

Density Analysis
Fixels weighted on fiber-bundle capacity and multi-d' organized

Association
Logistic Regression & ANN to find closest neighbor streamline.

Results

216% higher precision

11% higher recall

INTRODUCTION

White Matter Tractography: Non-invasive diffusion-imaging tool essential to analyzing brain matter connectivity through tractogram generation.[1]

- Preoperative surgical planning
  Most prevalent uses: Neoplastic lesions, electrical stimulation, language area surgery, vascular lesions, intracerebral hemorrhage, epilepsy, brainstem lesions, etc.
- Patient assessment and diagnosis
  Most prevalent uses: TBI assessment, stroke rehabilitation, psychiatric disorders, etc.
- Brain connectivity research and analysis
  Used in mapping connections between brain regions and understanding structural organization of neural circuits.

Key Nomenclature

Fixel	Directional peak in the probability distribution of a given fiber
Voxel	Similar to a pixel; a cubic segmentation of a dMRI image
Streamline	A coherent connected set of fixels constituting a white matter pathway
Bundle	A set of streamlines with similar initiations and terminations.

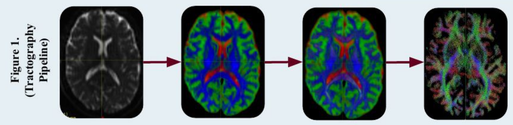


Figure 1. (Tractography Pipeline)

CURRENT DRAWBACKS

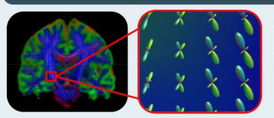


Figure 2. Sample of a full brain FOD.

Crossing fiber issue: False positives emerge when two streamlines cross due to a single prioritized peak vector.

- QBI and CSD utilizes probability distributions to isolate multiple peaks
- Number of other solutions to the issue

Bottleneck Issue: False positives emerge when two or more streamlines occur in a single voxel but cannot be resolved through angular resolution.[2]

- Imposing a voxel grid to the system creates partial volume effects that are dependent upon voxel size and fiber configuration.
- Leads to high rates of false-positive streamlines; current state-of-the-art algorithms have a ~**23%** precision score.

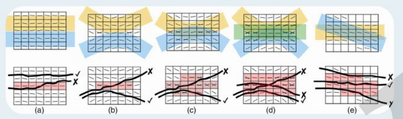
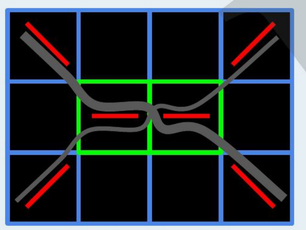
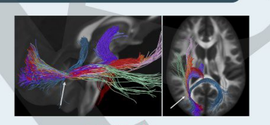


Figure 3. Visualization of the bottleneck issue in white matter tractography.

FUNDAMENTAL HYPOTHESIS





Legend
- Apparent Fixel Information
- Real Fixel Information
- Voxel-wise Bottleneck Location
- Adjacent Voxels
- Fiber Density Estimation

Figure 4. Visualization representing the basis of my fundamental hypothesis.

If current tractography results were deconstructed in specified bottleneck regions before reconstruction utilizing fiber density data, anatomical priors, and spatial information, this can solve the apparent false-positive paths formed when multiple fibers converge within limited angular resolution.

IDENTIFYING BOTTLENECKS

Defining A Bottleneck: A "multi-bundle fixel" in which multiple bundles aggregate within the same orientation of a voxel.[3]

FOD generation
mmt-CSD estimation of FOD using a group averaged WM response function.

Fixel Map
Peak generation with 0.1 probabilistic threshold to remove spurious peaks.

Tractography
Generation of a whole brain tractogram with FOD template.

Bundle recognition
Linear registration to HCP842 prior to bundle recognition by RecoBundlesX algorithm.

Bundle-Fixel TDI Map
Generation of a fixel tdi map through comparison with existing fixel image data.

Thresholding
TDI map is thresholded to create binary segmentations of fixel associations.

Indexing
Concurring fixels are labeled for bundle association and queued for extraction.

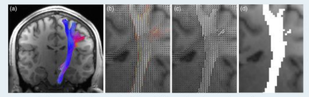


Figure 5. Steps in identifying bottleneck regions

Fiber Bundle Capacity (FBC)[4]: The total intra axonal fiber cross sectional area of a fiber bundle

- A measure of connectivity at an ROI of a given bundle.

Optimize $weight_s$ such that...
- $weight_s \times streamlines = measured\ FD$

Calculate proportionality coefficient …
- $\mu_{i,adj} = (AFD_{ref} / DWI_{ref}) \times |xyz| \times (\sum_{f \in i} (FD) / \sum_{f \in i} (TD))$

Therefore...
- $FBC = \mu_{i,adj} \times \sum_{s \in p} (weight_s)$

Developing Bottleneck Representations: Using each bottleneck voxel as a kernel, backpropagation to determine surrounding input "multi-bundle fixels"

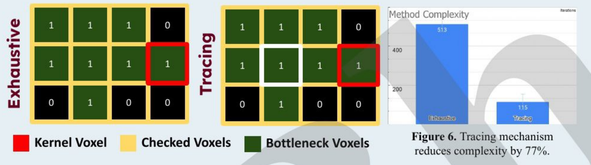
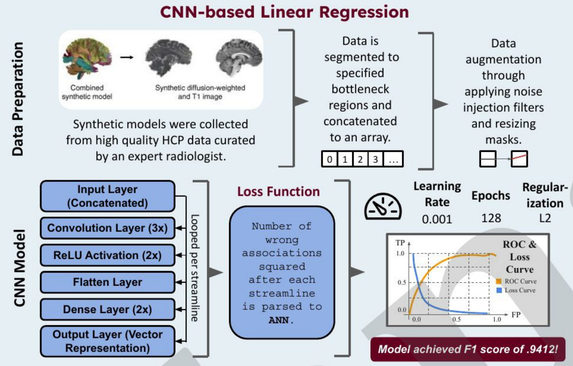


Figure 6. Tracing mechanism reduces complexity by 77%.

RESOLVING BOTTLENECKS

CNN-based Linear Regression



Approximate Nearest Neighbors: Algorithms to rapidly iterate through datasets and determine approximate neighbors of each point in a multi-dimensional space.

- **Locality Sensitive Hashing (LSH):** separates points into buckets of similarity.

Localized Streamline Reconstruction Once in/out streamlines are associated, localized streamline generation occurs. If streamline hits the end of a bottleneck region, it retraces and continues until it reaches the identified destination.

Dimension Reduction
Multiple randomized LSH functions split dataset to bins in low-dimensional space.

Indexing
Similar data points are mapped to the same bucket.

Query
Point projected to low dimensional space. Neighbor is identified through closeness in various bins.

Key Developmental APIs: MRtrix3, RecobundlesX, PyTorch, FAISS, DiPy

PERFORMANCE VALIDATION

5-step Verification[2]

1. Identification through Alternative Definition of "Valid Connection" (VC)
   (a) Bundle Clustering
   (b) Robust Shape
2. Remaining classified into "Invalid Connection" (IC)
3. ICs clustered through RecobundlesX
4. Best-matching endpoint regions for each cluster identified w/ majority voting.
5. Similar-termination bundles are merged.

VC ratio (Recall) TP/ TP + FN **94.52%**
Precision: TP/(TP + FP) **72.83%**

OL: # of traversed voxels within a ground truth bundle / total #voxels **79%**
OR: # of traversed voxels outside ground truth bundle / total #voxels **19%**

Tested on Synthetic HCP-derived validation dataset curated by expert radiologist.

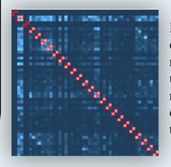


Figure 7. A connectivity matrix was utilized for normalized cross comparison across tractograms.

*TractX demonstrated a **216.15%** gain in precision from state-of-the-art algorithms! [2]*

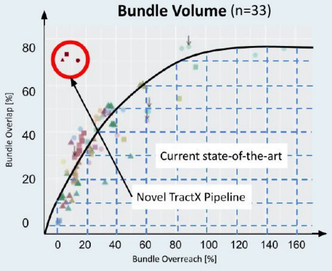
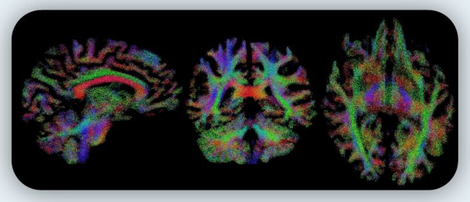


Figure 8. Graph of bundle overlap vs. bundle overreach. Huge boost in precision due to removal of bundle overreach and false-positives, leading to much higher results than the current state of the art.

Ground Truth \ Predicted Class	VC	IC
VC	True Positive 72.84%	False Positive 27.16%
IC	False Negative 3.42%	True Negative 96.58%

Figure 9. Normalized confusion matrix representing streamline distribution.

THE FIRST HIGH-PRECISION TRACTOGRAM



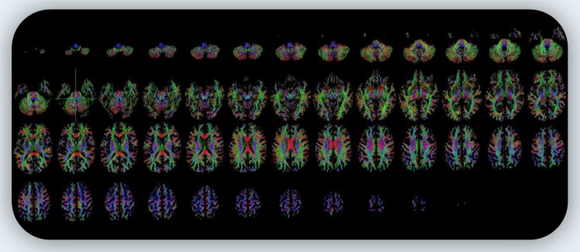


Figure 10 & 11. The first high-precision tractogram generated through the TractX pipeline, performing significantly higher with precision compared to other state of the art algorithm. Figure 10: Ortho View; Figure 11: Lightbox View; FSLeyes visualization.

LIMITATIONS AND SETBACKS

- Testing only completed on an M1-based machine.
- Require preset ground truth bundles to be defined for training.
- Prototype takes approximately ~14 hours to run complete pipeline.
- Performs significantly better (20% - 30%) with higher quality dMRI images (Optimized for 1mm).

- Registering and Transforming image data to retain scanner coordinates and relevant metadata.
- Constructing the tracing design to delineate bottlenecks due to hardware constraints.
- Optimizing the ANN to act as a loss function for the CNN in a conjoined network.
- Acquiring ground truth tractography data from various sources.

CONCLUSIONS

Invented tractography pipeline that drastically increases precision.

- Augments precision from 23% in state-of-the-art algorithms all the way up to 72.83%, a change of **216.15%**.
- Minimizes false-positive streamline detection, making tractography a much **more reliable tool for research** and analysis.

Provides enhanced tools to surgical settings.

- **Powerful pre-operative planning tool**: Unlock better surgical outcomes for millions with powerful insights, ultimately leading to better patient care.
- Accessible and user-friendly: A **command-line interface** that can easily run on common computer systems and doesn't require extensive radiological expertise.

Designed a fiber density analysis system.

- Beyond Connectivity: Leveraging fiber density for advanced analysis opens the door to more exciting applications in fiber **analysis and researching neural circuits**.

Novel conjoined network between a CNN and ANN.

- Versatile Architecture: Could have wide usage across various fields such as **astronomy, image interpreters, and signal filtration**.

Further improvements.

- Utilizing greater diversity for anatomical priors to deal with further corner cases.
- Integration with microstructural information to determine most likely path.

KEY REFERENCES

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Figure 3, 5. (Schilling et al.)

All other information and graphics were created by the researcher (Krishna Bhatt).